# Escape Dynamics in the Hénon–Heiles System

Numerical investigation of trajectories, escape channels, and sensitivity to initial conditions

**Amogh Nataraj** – EP25BTECH11006
**Premkumar Gandhi** – EP25BTECH11010
**Pritam Bagle** – EP25BTECH11015
Indian Institute of Technology Hyderabad
Computational Physics – EP4210

**GitHub repository:** github.com/mogit5756/Henon-Heiles-Problem-Implementation
**Notebook:** `Henon_Heiles_Python_Implementation.ipynb`

**Abstract**

The Hénon–Heiles system is a two-dimensional Hamiltonian model with both bounded trajectories and escape through three equivalent channels. In this project, we examine how the launch angle and total energy affect which channel a particle reaches and how long the escape takes. We implemented the main simulations using the Velocity Verlet method, checked energy conservation as the timestep was reduced, and used SciPy's DOP853 solver for selected comparisons and Poincaré sections. At $E = 0.18$, 228 of the 360 sampled launch angles escaped within $T = 100$, with equal numbers using each channel. The energy scan showed a larger finite-time escape fraction at higher energies. A finer scan near one angular boundary also showed that some predicted outcomes change with numerical resolution. Overall, the simulations show how energy and launch direction affect escape, and where numerical resolution makes individual predictions less certain.

## 1 Introduction

The Hénon–Heiles system was introduced in the study of stellar dynamics as a simple model for motion in a non-integrable gravitational potential [1]. It has only two degrees of freedom and a conserved total energy, yet its nonlinear equations can produce very different types of motion. Depending on the energy and the initial condition, a particle may stay in the central region or leave it through one of three equivalent escape channels.

Our main question is: *when the total energy is fixed, how does the launch direction affect the escape channel and escape time? How do those outcomes change when the energy or launch angle is varied?* Although the total energy remains constant, the particle's trajectory and escape channel still depend on its initial conditions. The Hénon–Heiles system is non-integrable for general initial conditions: energy is the conserved quantity available for the general motion, but there is no second independent global integral that reduces every trajectory to a closed-form solution. Numerical integration is therefore needed to follow individual orbits and compare many launches under controlled conditions.

We first study individual trajectories and their energy error. We then scan the launch angle, vary the initial energy, and examine Poincaré sections. Finally, we look closely at a narrow range of launch angles to see how the predicted outcomes change with angular and integration resolution. The energy and resolution scans provide the parameter studies that extend the basic trajectory calculation. Throughout the report, we distinguish between a particle that has not escaped during the simulated time and one that is permanently trapped; our calculations can establish only the former.

## 2 Physical model

We use dimensionless coordinates $(x, y)$ and take the particle mass to be one. The potential is

$$V(x, y) = \frac{1}{2}(x^2 + y^2) + x^2 y - \frac{1}{3}y^3. \tag{1}$$

The Hamiltonian, equal to the total mechanical energy, is

$$H = E = \frac{1}{2}(v_x^2 + v_y^2) + V(x, y), \tag{2}$$

where $v_x = \dot{x}$ and $v_y = \dot{y}$. Since the potential is time independent, the exact dynamics conserve $E$.

For unit mass, Newton's second law gives $\ddot{\boldsymbol{r}} = -\nabla V$. Differentiating the potential gives

$$\frac{\partial V}{\partial x} = x + 2xy, \qquad \frac{\partial V}{\partial y} = y + x^2 - y^2. \tag{3}$$

The resulting equations of motion are

$$\boxed{\ddot{x} = -x - 2xy, \qquad \ddot{y} = -y - x^2 + y^2.} \tag{4}$$

For numerical integration with a general ODE solver, the same model can be written as four first-order equations:

$$\dot{x} = v_x, \qquad \dot{y} = v_y, \qquad \dot{v}_x = -x - 2xy, \qquad \dot{v}_y = -y - x^2 + y^2. \tag{5}$$

The exact equations conserve the total energy. This can be checked directly by differentiating Eq. (2) with respect to time:

$$\frac{dE}{dt} = v_x\dot{v}_x + v_y\dot{v}_y + \frac{\partial V}{\partial x}\dot{x} + \frac{\partial V}{\partial y}\dot{y} \tag{6}$$

$$= v_x\left(\dot{v}_x + \frac{\partial V}{\partial x}\right) + v_y\left(\dot{v}_y + \frac{\partial V}{\partial y}\right) = 0, \tag{7}$$

where $\dot{x} = v_x$, $\dot{y} = v_y$, and Newton's equations imply $\dot{v}_x = -\partial V/\partial x$ and $\dot{v}_y = -\partial V/\partial y$. Numerical energy drift is consequently a measure of integration error.

The escape threshold is set by the saddle points of the potential. To find them, we set both components of the gradient to zero:

$$x(1 + 2y) = 0, \qquad y + x^2 - y^2 = 0. \tag{8}$$

There are two cases. If $x = 0$, the second equation gives $y = 0$ or $y = 1$. The point $(0, 0)$ is the central minimum. If $x \neq 0$, the first equation requires $y = -1/2$; substituting this into the second gives $x^2 = 3/4$. The three non-central stationary points are therefore

$$(0, 1), \qquad \left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right), \qquad \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right). \tag{9}$$

At the origin, the Hessian is positive definite, so the origin is a local minimum. At each of the other three points, the Hessian

$$\nabla^2 V = \begin{pmatrix} 1 + 2y & 2x \\ 2x & 1 - 2y \end{pmatrix} \tag{10}$$

has determinant $-3$, confirming that each is a saddle. By symmetry, all three have the same potential energy. Evaluating the potential at $(0, 1)$ gives

$$E_c = V(0, 1) = \frac{1}{2} - \frac{1}{3} = \boxed{\frac{1}{6} \simeq 0.1667}. \tag{11}$$

The other two saddles have the same energy because the potential has threefold rotational symmetry. Below $E_c$, the allowed region around the origin is closed. Above $E_c$, the three escape passages are open. This makes escape possible, but it does not mean that every initial condition leads to escape within the time simulated.

# 3 Numerical methodology

## 3.1 Initial conditions and integration

For the angular scans, every particle starts at the origin. We keep the total energy fixed and change only the direction of the initial velocity. Since $V(0, 0) = 0$, the initial conditions for a launch angle $\theta$ are

$$x_0 = y_0 = 0, \qquad v_{x,0} = \sqrt{2E}\cos\theta, \qquad v_{y,0} = \sqrt{2E}\sin\theta. \tag{12}$$

These expressions come from the energy constraint. At the origin, $V(0, 0) = 0$, and hence

$$E = \frac{1}{2}(v_{x,0}^2 + v_{y,0}^2), \qquad v_0 = \sqrt{v_{x,0}^2 + v_{y,0}^2} = \sqrt{2E}. \tag{13}$$

Resolving this velocity vector into its $x$ and $y$ components gives Eq. (12). Therefore, a fixed-energy angular scan compares launches with the same initial speed; only their directions differ.

We use a custom implementation of the second-order Velocity Verlet method for the main trajectory calculations. Let $\boldsymbol{q} = (x, y)$ denote position, $\boldsymbol{v} = (v_x, v_y)$ velocity, and $\boldsymbol{a}(\boldsymbol{q}) = -\nabla V(\boldsymbol{q})$ acceleration. One step of size $h$ is

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{v}_n + \frac{h^2}{2}\boldsymbol{a}(\boldsymbol{q}_n), \tag{14}$$

$$\boldsymbol{v}_{n+1} = \boldsymbol{v}_n + \frac{h}{2}\left[\boldsymbol{a}(\boldsymbol{q}_n) + \boldsymbol{a}(\boldsymbol{q}_{n+1})\right]. \tag{15}$$

The position update comes from keeping terms through second order in the Taylor expansion

$$\boldsymbol{q}(t+h) = \boldsymbol{q}(t) + h\boldsymbol{v}(t) + \frac{h^2}{2}\boldsymbol{a}(t) + O(h^3). \tag{16}$$

For the velocity, integrate $d\boldsymbol{v}/dt = \boldsymbol{a}$ over one step and approximate the acceleration integral with the trapezoidal rule:

$$\boldsymbol{v}(t+h) = \boldsymbol{v}(t) + \int_t^{t+h} \boldsymbol{a}\, dt \simeq \boldsymbol{v}(t) + \frac{h}{2}\left[\boldsymbol{a}(t) + \boldsymbol{a}(t+h)\right]. \tag{17}$$

Because acceleration depends on position, we first calculate the new position and then evaluate the acceleration there. This gives Eq. (15). For this separable Hamiltonian, Velocity Verlet is time-reversible and symplectic, and its global error is second order in the timestep. The main loop used in the notebook can be summarized as follows:

**Listing 1:** Outline of one Velocity Verlet step and the escape check.

```
for each time step n:
    a_n = acceleration(q_n)
    q_(n+1) = q_n + h v_n + 0.5 h^2 a_n
    a_(n+1) = acceleration(q_(n+1))
    v_(n+1) = v_n + 0.5 h (a_n + a_(n+1))
    if the outward escape condition is met:
        record escape time and channel
        stop integration
```

## 3.2 Escape detection and channel assignment

We measure distance from the origin using $r = \sqrt{x^2 + y^2}$. Differentiating this expression gives the radial velocity:

$$v_r = \dot{r} = \frac{d}{dt}(x^2+y^2)^{1/2} = \frac{x\dot{x} + y\dot{y}}{r} = \frac{xv_x + yv_y}{r}, \qquad r > 0. \tag{18}$$

The condition $v_r > 0$ means that the particle is moving away from the origin. In the main simulations, we record an escape when a stored time step first satisfies $r > R$ and $v_r > 0$, with $R = 2.5$. We use the angle $\phi = \mathrm{atan2}(y, x)$ at that point to assign the exit channel. The code uses the sectors 60°–120° (upper), 180°–240° (lower-left), and 300°–360° (lower-right). These ranges are practical labels for the three exits, not additional physical boundaries. Since the test is made at discrete time steps, the reported Velocity Verlet escape time is resolved to the timestep. If a trajectory has not crossed the radius by the end of the run, we label it *not escaped by* $T$; we do not treat it as permanently trapped.

## 3.3 Independent solver and Poincaré sections

We use SciPy's `solve_ivp` with DOP853, an adaptive high-order Runge–Kutta solver [3], in two supporting tasks: generating the Poincaré sections and checking selected Velocity Verlet results. It is not used for the full angular or energy scans. The four first-order equations in Eq. (5) are supplied to the solver. For the representative solver comparison, we set `rtol` $= 10^{-12}$, `atol` $= 10^{-14}$, and a maximum step of 0.01. For the boundary-angle checks, the maximum step was tested at 0.005, 0.0025, and 0.00125, using the same tolerances. For a Poincaré section, we record crossings of $x = 0$ in the positive direction ($v_x > 0$) and plot the crossing coordinates $(y, v_y)$. The initial point at $t = 0$ is excluded.

For each Poincaré section, we use 15 equally spaced starting values of $y_0$ between $-0.30$ and 0.45, with $x_0 = 0$ and $v_{y,0} = 0$. The remaining initial velocity is chosen from the energy constraint. At $x = 0$, the potential becomes $V(0, y) = \frac{1}{2}y^2 - \frac{1}{3}y^3$, giving

$$v_{x,0} = +\sqrt{2\left[E - \left(\frac{1}{2}y_0^2 - \frac{1}{3}y_0^3\right)\right]}. \tag{19}$$

We use only initial values for which the square root is real. The sections are calculated at $E = 1/12$, $1/8$, $1/6$, and 0.18, with integration continued to $T = 1000$ or stopped earlier if the escape event occurs. The Poincaré calculation uses `rtol` $= 10^{-9}$ and `atol` $= 10^{-11}$. Points are coloured by their starting value $y_0$, so crossings from each of the 15 trajectories can be distinguished. The longer observation time gives each non-escaping trajectory more crossings and makes the section structure easier to see.

### 3.4 Quantities measured in the parameter scans

For a scan with $N_\theta$ launch angles at a fixed energy, we define the finite-time escape fraction as

$$f_{\mathrm{esc}}(E,T) = \frac{N_{\mathrm{esc}}(E,T)}{N_\theta}, \tag{20}$$

Here, $N_{\mathrm{esc}}$ is the number of trajectories that meet the outward escape condition before time $T$. We calculate the median escape time using only trajectories that escaped. To measure energy conservation, we record the largest absolute deviation from the initial energy:

$$\Delta E_{\max} = \max_n |E_n - E_0|, \tag{21}$$

with the maximum taken over the stored trajectory up to escape or the final integration time.

We also compare nearby launch directions. Let $C_h(\theta)$ be the outcome label at timestep $h$, with "not escaped by $T$" treated as its own outcome. If the sampled angles are separated by $\Delta\theta$ and we compare pairs separated by $\delta\theta = m\Delta\theta$, the fraction of pairs with different labels is

$$f_{\mathrm{change}}(\delta\theta; h) = \frac{1}{N_\theta - m} \sum_{i=0}^{N_\theta - m - 1} \mathbb{I}[C_h(\theta_i) \neq C_h(\theta_{i+m})], \tag{22}$$

The indicator $\mathbb{I}$ equals one when the two outcomes differ and zero otherwise. The resulting curve gives a finite-grid measure of how sensitive the sampled outcomes are to changes in launch direction.

### 3.5 Main numerical settings

**Table 1:** Settings used for the principal numerical experiments.

| Experiment | Settings | Purpose |
|---|---|---|
| Confined energy test | $E = 0.08$, $T = 100$, $\theta = 20°$, $N$ = 5,000–20,000 | Energy error |
| Escaping energy test | $E = 0.18$, $T = 100$, $\theta = 90°$, $N$ = 10,000–40,000 | Energy error |
| Symmetry test | $E = 0.18$, $T = 100$, $N$ = 10,000; 90°, 210°, 330° | Symmetry |
| Full angular scan | $E = 0.18$, $T = 100$, $N$ = 20,000, 1° spacing | Channel and time maps |
| Coarse energy scan | $E$ = 0.15–0.20, $N$ = 20,000, 10° spacing | Energy dependence |
| Refined energy scan | $E$ = 0.167–0.20, $N$ = 20,000, 5° spacing | Near-threshold trend |
| Local angular scan | 22°–23°, 0.01° spacing; $N$ = 20,000, 40,000 | Resolution sensitivity |
| Poincaré sections | $T = 1000$, 15 starts per energy, $R = 2.5$; DOP853, `rtol` $= 10^{-9}$, `atol` $= 10^{-11}$ | Phase-space sections |

For Velocity Verlet, the timestep is $h = T/N$. The main angular and energy scans therefore use $h = 0.005$. The other settings in the table are stated explicitly; the escape radius is $R = 2.5$ unless noted otherwise.

## 4 Numerical validation

### 4.1 Energy conservation and timestep scaling

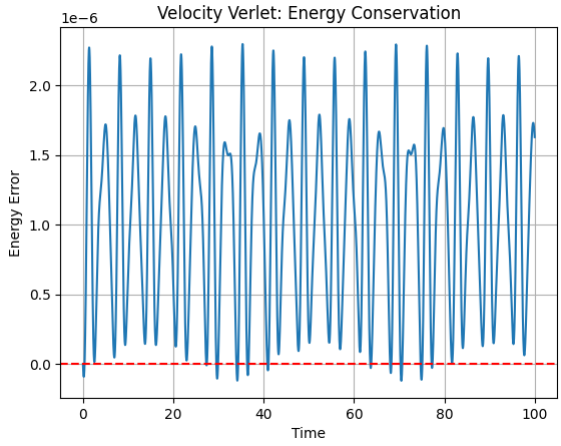
For each run, we calculate $|E_n - E_0|$ at every stored time step and take its maximum. For the below-threshold trajectory ($E = 0.08$, $\theta = 20°$), the maximum error falls from $9.19 \times 10^{-6}$ at $N = 5{,}000$ to $2.30 \times 10^{-6}$ at $N = 10{,}000$, and then to $5.74 \times 10^{-7}$ at $N = 20{,}000$. Each time the number of steps is doubled, the error is reduced by about a factor of four, as expected for a second-order method.

The same pattern appears in the escaping trajectory at $E = 0.18$ and $\theta = 90°$. Its maximum error decreases from $2.17 \times 10^{-4}$ at $N = 10{,}000$ to $5.42 \times 10^{-5}$ at $N = 20{,}000$ and $1.34 \times 10^{-5}$ at $N = 40{,}000$. The errors are larger in this run, but the reduction with timestep is again close to a factor of four.

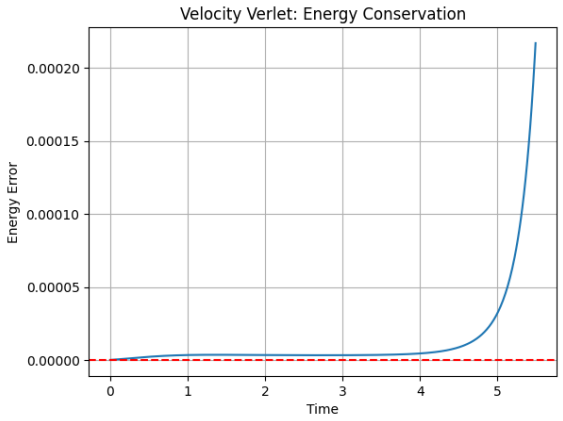
### 4.2 Symmetry and independent solver comparison

The potential is unchanged by a rotation of 120°. We tested this by rotating the trajectory launched at 90° and comparing it with the trajectories launched at 210° and 330°. The largest coordinate differences were around $10^{-14}$, close to floating-point precision. This is a useful check that the implementation respects the expected threefold symmetry.

We also compared the two solvers for a representative launch at $E = 0.18$ and $\theta = 90°$. DOP853 recorded an escape time of 5.48554, while Velocity Verlet with $N = 20{,}000$ gave 5.49, a difference of 0.00446. Up to the




**(a)** $E = 0.08$, confined trajectory.




**(b)** $E = 0.18$, upper-channel launch.

**Figure 1:** Energy error histories for a below-threshold and an above-threshold trajectory. The maximum errors and their timestep dependence are reported in the text.

DOP853 escape time, the largest difference in position was $1.05 \times 10^{-4}$. The maximum DOP853 energy error was $2.28 \times 10^{-15}$. For this trajectory, the two methods agree closely.

The two methods do not agree as well near the angular basin boundary. For 22.05° and 22.46°, DOP853 gave the same channel outcome when its maximum step was reduced from 0.005 to 0.00125, but Velocity Verlet was more sensitive to its timestep. At 22.46°, the finest Velocity Verlet run ($N = 640{,}000$) had not escaped by $T = 100$, whereas DOP853 predicted an upper-channel escape at about 97.2945. At 22.05°, both methods predicted an upper-channel escape, but their escape times still differed: about 85.433 for Velocity Verlet and 93.145 for DOP853. Small differences between numerical trajectories can grow during a long integration. Near an escape-basin boundary, this can be enough to change the final channel or whether escape is recorded before $T = 100$. The DOP853 step-size test is reassuring, but it does not make that solver exact; neither result should be treated as the unquestionable answer. These cases show why individual outcomes near the boundary need to be interpreted with care.

# 5 Results and analysis

## 5.1 Trajectories in the potential

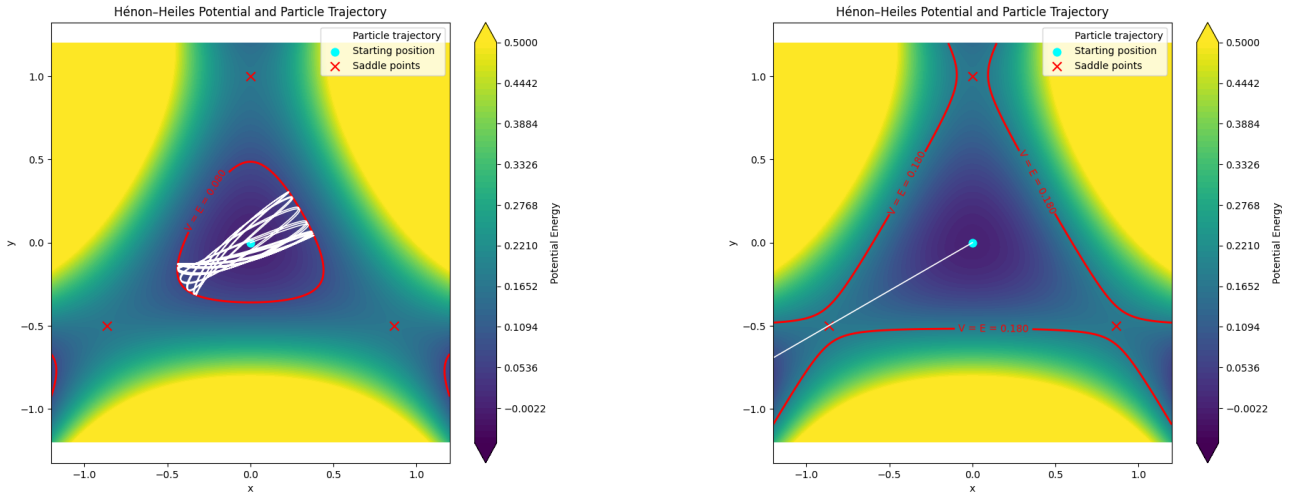
Figure 2 shows one trajectory below the escape threshold and one above it. At $E = 0.08$, the allowed region around the origin is closed and the trajectory remains bounded throughout the $T = 100$ run. At $E = 0.18$, the energy is above the saddle value and the three escape passages are open. The plotted lower-left trajectory moves towards one of these passages; the code records escape only when it crosses the chosen radius while moving outwards. These examples show how the accessible region changes at the threshold, but they do not establish the long-term fate of every initial condition.

## 5.2 Escape channel and escape time versus launch angle

For the full angular scan at $E = 0.18$, we used 360 launch angles from 0° to 359°, separated by 1°. Each trajectory was integrated to $T = 100$ using $N = 20{,}000$ steps and an escape radius of $R = 2.5$. Of the 360 launches, 228 (63.3%) escaped before the end of the run and 132 did not. The escaped trajectories were divided equally among the channels: 76 upper, 76 lower-left, and 76 lower-right. No escape was left unclassified. This equal count is consistent with the threefold symmetry of the potential and the uniform angular grid.

For the 228 trajectories that escaped, the mean escape time was 29.86 and the median was 14.9. The shortest recorded time was 5.49 and the longest was 98.115. Figure 3 shows the channel and escape time for each launch angle. Crosses placed at $T = 100$ mark launches that did not escape during the run; they are not assigned an infinite escape time. The full circular scan also contained 144 changes between neighbouring outcome labels at the 1° spacing, counting "not escaped" as a separate outcome and including the comparison between 359° and 0°. This count depends on the angular grid and is used only to describe the sampled pattern.

The shortest recorded escape time in this scan is 5.49, occurring for the axial launch at 90° and, by threefold symmetry, for the corresponding launches at 210° and 330°. For the 90° launch, $x_0 = v_{x,0} = 0$ and the equation $\ddot{x} = -x(1 + 2y)$ keeps the motion on the line $x = 0$. The problem then reduces to one-dimensional motion with



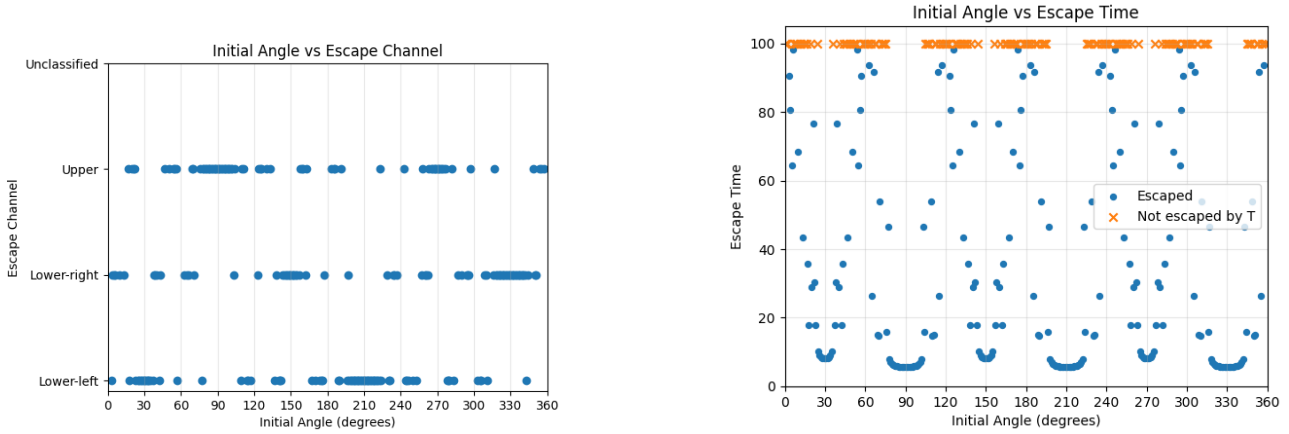

**(a)** Below threshold: $E = 0.08$. **(b)** Above threshold: $E = 0.18$.

**Figure 2:** Trajectories overlaid on the potential-energy contours. Red crosses mark the three saddle points, and the red contour marks $V = E$.

$\ddot{y} = -y + y^2$ and $V(0, y) = \frac{1}{2}y^2 - \frac{1}{3}y^3$. The continuous escape time to $y = R = 2.5$ is

$$t_{\text{esc}} = \int_0^{2.5} \frac{dy}{\sqrt{2\left[E - \left(\frac{1}{2}y^2 - \frac{1}{3}y^3\right)\right]}}. \tag{23}$$

At $E = 0.18$, this integral gives $t_{\text{esc}} \simeq 5.48554$. The Velocity Verlet scan reports 5.49 because its escape check is made at discrete time steps of $h = 0.005$.




**(a)** Exit channel. **(b)** Escape time.

**Figure 3:** Angular dependence of the escape outcome at $E = 0.18$, sampled at 1° spacing (360 launches) and integrated to $T = 100$. Crosses mark launches that did not escape within this observation time.

## 5.3 Poincaré sections

A Poincaré section summarizes a continuous trajectory by recording where it crosses a chosen surface in phase space. Here, Fig. 4 shows successive crossings of $x = 0$ with $v_x > 0$, plotted in the $(y, v_y)$ plane. With 15 initial conditions at each energy and an observation time of $T = 1000$, the regenerated calculation recorded 2370, 2355, 2307, and 844 crossings at $E = 1/12$, $1/8$, $1/6$, and 0.18, respectively. No trajectories crossed the escape radius at the first three energies. At $E = 0.18$, 11 of the 15 trajectories escaped before $T = 1000$. Since integration stops at the escape event, these early exits contribute fewer later crossings, which explains why this panel is much sparser.

The colour scale identifies the starting value $y_0$ of each trajectory. At $E = 1/12$, the sampled trajectories form smooth, organized curves, characteristic of regular motion. The $E = 1/8$ section is spread over a broader region. At $E = 1/6$, a compact two-lobed island is visible near the centre, surrounded by a broader distribution of crossings. This structure is consistent with regular and more irregular motion coexisting among the sampled trajectories. The $E = 0.18$ panel is sparse because 11 trajectories escape before the end of the integration, so

their crossings stop accumulating. These sections are qualitative descriptions of the selected initial conditions and do not classify the full phase space.

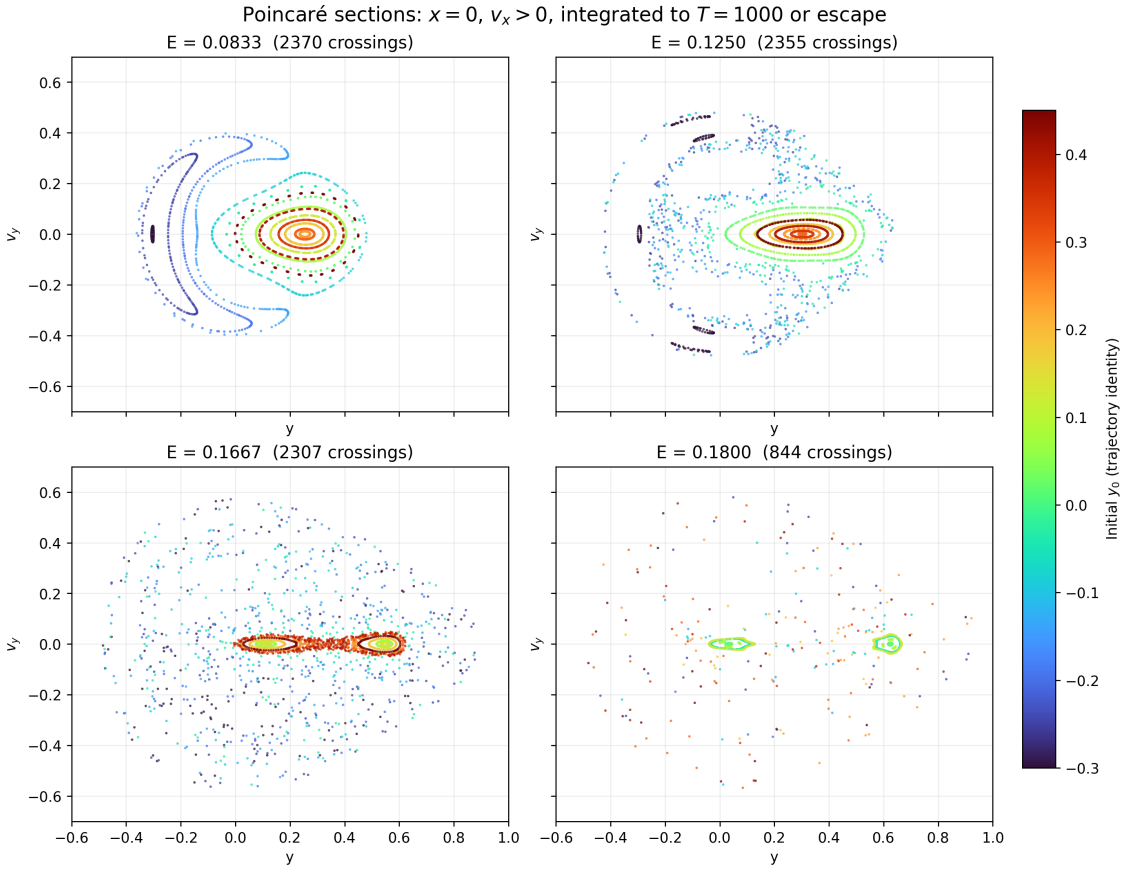



**Figure 4:** Poincaré sections at four energies, using $x = 0$ and positive crossings $v_x > 0$. Colours identify the starting value $y_0$. Each orbit is integrated to $T = 1000$ or until escape. The displayed window is $y \in [-0.6, 1.0]$ and $v_y \in [-0.7, 0.7]$; three isolated crossings outside this window in the $E = 0.18$ panel are omitted from view but included in the crossing count.

## 5.4 Dependence of escape fraction on energy

We measured the escape fraction at several energies using uniform launch-angle grids, with $N = 20{,}000$, $T = 100$, and $R = 2.5$. The coarse scan used 10° spacing, while the refined scan used 5° spacing. The refined scan gave 6 escapes out of 72 launches at $E = 0.167$, 18 at $E = 0.170$, 24 at $E = 0.175$, 54 at $E = 0.18$, and all 72 at $E = 0.20$. The coarse scan also included two below-threshold cases, $E = 0.15$ and 0.165, for which no launches escaped. Table 2 compares the two grids, and Fig. 5 shows the refined results.

**Table 2:** Finite-time escape counts at $T = 100$ for the coarse (10°, 36 launches) and refined (5°, 72 launches) angular grids, using $N = 20{,}000$ steps per trajectory. Each entry gives the number escaped and the percentage of sampled launches. A dash indicates that the energy was not included in that scan.

| $E$ | 10° grid: escaped/36 | 5° grid: escaped/72 |
|---|---|---|
| 0.150 | 0 (0.0%) | – |
| 0.165 | 0 (0.0%) | – |
| 0.167 | – | 6 (8.3%) |
| 0.170 | 6 (16.7%) | 18 (25.0%) |
| 0.175 | – | 24 (33.3%) |
| 0.180 | 30 (83.3%) | 54 (75.0%) |
| 0.200 | 36 (100%) | 72 (100%) |

The overall trend is clear: at higher energies, a larger fraction of the sampled directions escape within $T = 100$. The measured fraction at $E = 0.18$ depends noticeably on the angular grid: the 1° scan gives $228/360 = 0.633$, the 5° scan gives $54/72 = 0.750$, and the 10° scan gives $30/36 = 0.833$. This difference can be understood directly from how the deterministic grids are constructed. The 5° grid samples the 72 angles $0°, 5°, \ldots, 355°$, which are also present in the 1° scan; 54 of these 72 angles escape. The other 288 angles in the 1° scan contain 174 escapes, or 60.4%. Thus, the every-fifth-degree subset happens to select a larger share of escaping directions than the four

intervening one-degree classes. The 10° grid is a further subset of the 5° grid and similarly gives $30/36 = 83.3\%$. Because the escape outcomes vary non-uniformly with launch angle and can change over narrow angular intervals, a regular grid can over- or under-sample those intervals depending on its spacing and alignment. These values are therefore deterministic finite-grid fractions, not independent statistical estimates or exact escape probabilities. The 1° grid is the densest of the three, but its result is still conditional on the finite observation time $T = 100$.

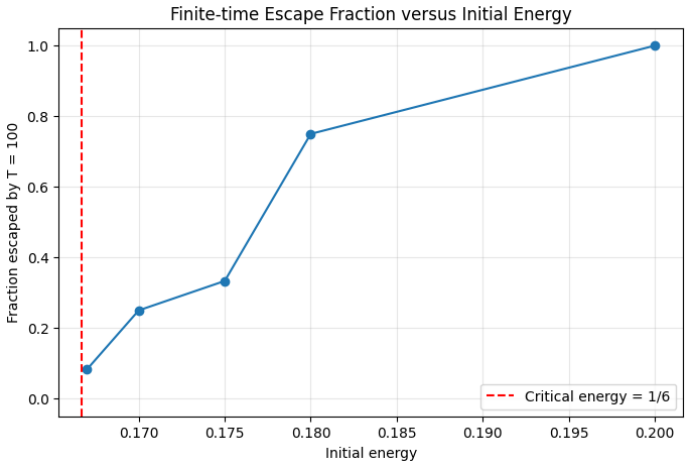
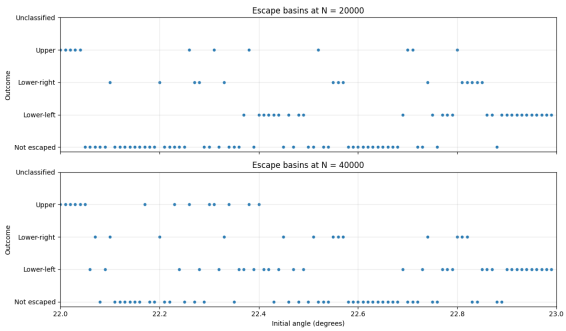



**Figure 5:** Finite-time escape fraction from the refined energy scan, using 5° angular spacing (72 launches per energy) and observation time $T = 100$. The vertical line marks the theoretical threshold $E_c = 1/6$.

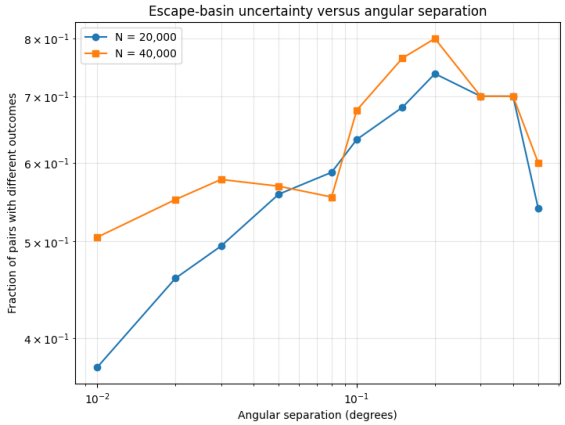
## 5.5 Sensitivity near an angular basin boundary

We then focused on the interval from 22° to 23°, where neighbouring angles often lead to different outcomes. The scan contains 100 angles at 0.01° spacing. The number of changes between adjacent labels was 37 for $N = 20{,}000$ and 50 for $N = 40{,}000$. When the two timestep resolutions were compared point by point, 31 of the 100 launch angles (31%) changed their label. The local outcome map is therefore sensitive to the integration timestep in this interval.

We also compared pairs of launches separated by $\delta\theta$ between 0.01° and 0.50°. For example, at 0.01°, the outcome-change fractions were 0.374 and 0.505 for $N = 20{,}000$ and 40,000, respectively; at 0.20°, they were 0.738 and 0.800. The values do not increase at every point, and the two resolutions do not give identical curves. We therefore use these curves only as exploratory, finite-grid sensitivity diagnostics.




**(a)** Outcome labels at two timestep resolutions.




**(b)** Fraction of pairs with different outcomes.

**Figure 6:** Outcome labels at two timestep resolutions and the fraction of nearby angle pairs with different outcomes in the 22°–23° interval at $E = 0.18$.

# 6 Discussion and limitations

The saddle energy determines when the three escape passages become accessible. Above this threshold, the launch angle still matters: it changes both the channel reached and the time taken to reach it. At $E = 0.18$, the full angular scan gives equal channel counts, while the energy scans show a larger finite-time escape fraction at higher energies. Together, these results show why energy alone is not enough to predict an individual trajectory.

The main advantage of Velocity Verlet is that it is straightforward to implement and, for this Hamiltonian system, preserves the symplectic structure. The timestep tests also show good energy behaviour for the cases examined. There are still limits to what we can conclude. We define escape using a chosen radius $R = 2.5$, an outward crossing, and fixed angular sectors for the three channels. All angular scans launch particles from the origin and use finite grids; the Poincaré sections use a more restricted set of initial conditions. Each simulation also ends at a chosen final time, so "not escaped by $T$" does not mean that a particle is trapped forever. Near the 22°–23° boundary, some classifications change with timestep, and one tested case still disagrees with DOP853. Velocity Verlet uses a fixed timestep, so it cannot automatically take smaller steps near these sensitive trajectories.

An escape basin is the set of starting conditions that lead to a particular exit channel. Our fine scan shows an irregular boundary between the sampled outcomes, but that pattern alone does not prove that the boundary is fractal. A fractal dimension measures how the detail of a set changes as the scale of observation is reduced; fractal sets can have non-integer dimensions. The Wada property is a stronger condition: every point on the boundary of one escape basin must also lie on the boundaries of the other two. This means that initial conditions arbitrarily close to such a boundary point can lead to all three escape channels. We have not calculated a fractal dimension or tested the Wada property, so we make no claim that either is present [2].

The Poincaré plots also use a limited family of initial conditions and a finite integration time, so they are qualitative rather than a classification of the full phase space. Further work could sample a wider range of initial conditions and repeat the boundary tests with finer angular and temporal resolution.

# 7 Conclusion

Our simulations show that both energy and launch direction affect escape in the Hénon–Heiles system. The energy-error and symmetry tests support the implementation for the cases examined. The energy scans show a generally larger finite-time escape fraction at higher energies, although the measured fraction depends on the angular grid. The Poincaré sections show organized curves at low energy and a two-lobed island near $E = 1/6$. For the 90° axial launch, the one-dimensional calculation predicts an escape time of 5.48554, close to the Velocity Verlet value of 5.49. The fine angular scan also shows that predictions close to an escape boundary can change with the timestep.

# References

[1] M. Hénon and C. Heiles, "The Applicability of the Third Integral of Motion: Some Numerical Experiments," *The Astronomical Journal*, vol. 69, pp. 73–79, 1964. doi:10.1086/109234.

[2] J. Aguirre, J. C. Vallejo, and M. A. F. Sanjuán, "Wada basins and chaotic invariant sets in the Hénon–Heiles system," *Physical Review E*, vol. 64, 066208, 2001. doi:10.1103/PhysRevE.64.066208.

[3] SciPy documentation, "`scipy.integrate.solve_ivp`," https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html.